

Riemann Hypothesis as Low-Frequency Annihilation in a Mirror–Odd Boundary Channel: A Reduction and a Verified Spectral Gap in the Ordinates-Only Model

Jan Mikulík

January 16, 2026

Abstract

We study a mirror–odd boundary trace built from the logarithmic derivative of the completed Riemann xi-function in a trimmed strip. After (i) mirror–odd projection, (ii) a smooth dipole (weighted mean) subtraction, and (iii) time smoothing, the Riemann Hypothesis (RH) is reduced to a single low-frequency energy inequality for an explicit filtered boundary channel. The reduction is deterministic and uses only classical harmonic analysis (Poisson control from boundary to interior) and a structural Taylor annihilation lemma induced by the dipole filter (DC-kill). The remaining analytic bottleneck is an explicit safe-line transfer identity expressing the filtered boundary channel as a bounded linear transform of ζ'/ζ on a safe line $\Re s = \frac{1}{2} + \varepsilon$ plus a controlled remainder. We complement the reduction with numerical certification in an ordinates-only model: at height $t_0 \approx 2.6765 \times 10^{11}$ using $N = 10^4$ zeros, the measured low-frequency energy decays as $F_T \asymp T^{-\hat{\beta}}$ with $\hat{\beta} \approx 1.786$ (margin $\hat{\beta} - 1 \approx 0.786$), and on the upper half of the T -range $\hat{\beta} \approx 2.97$ (margin ≈ 1.97).

1 Setup in a trimmed strip

Let $\xi(s)$ be the completed xi-function and define

$$S(\sigma, t) := \partial_\sigma \log |\xi(\sigma + it)|.$$

Fix $\varepsilon \in (0, \frac{1}{2})$ and $\delta \in (0, \frac{1}{2} - \varepsilon)$ and work in the trimmed strip

$$I_{\varepsilon, \delta} := [\varepsilon + \delta, 1 - \varepsilon - \delta].$$

On $I_{\varepsilon, \delta}$ we write the standard decomposition

$$S(\sigma, t) = A(t) \left(\sigma - \frac{1}{2} \right) + E(\sigma, t), \quad A(t) = \frac{1}{2} \log \frac{t}{2\pi} + O(1),$$

where $E(\sigma, t)$ is harmonic in σ for each fixed t .

2 Mirror–odd boundary channel and smoothing

Define the mirror–odd projection in σ :

$$e_{\text{odd}}(\sigma, t) := \frac{E(\sigma, t) - E(1 - \sigma, t)}{2}.$$

Fix an even Schwartz window W with $\int_{\mathbb{R}} W(u) du = 1$ and define the t -smoothing

$$(W_T * f)(t) := \int_{\mathbb{R}} f(t+u) \frac{1}{T} W\left(\frac{u}{T}\right) du, \quad T = T(t) \rightarrow \infty \text{ slowly.}$$

We study the smoothed boundary derivative

$$g(t) := (W_T * \partial_{\sigma} e_{\text{odd}}(\varepsilon, \cdot))(t).$$

3 The smooth dipole (weighted mean subtraction)

Let $w \in C_c^{\infty}(\mathbb{R})$ be even, nonnegative, $\int w = 1$. Define $w_H(u) = H^{-1}w(u/H)$ and the mean operator

$$(\mathcal{M}_H f)(t) := (w_H * f)(t), \quad \mathcal{D}_H := I - \mathcal{M}_H.$$

We define the dipole-filtered boundary signal

$$g_H(t) := (\mathcal{D}_H g)(t).$$

Lemma 1 (Convolution dipole projector: L^2 and H^1 stability). *For all $f \in L^2(\mathbb{R})$,*

$$\|\mathcal{D}_H f\|_{L^2} \leq \|f\|_{L^2}.$$

If moreover $f \in H^1(\mathbb{R})$, then

$$\|(\mathcal{D}_H f)'\|_{L^2} \leq \|f'\|_{L^2}.$$

Proof. Young's inequality gives $\|\mathcal{M}_H f\|_2 \leq \|w_H\|_1 \|f\|_2 = \|f\|_2$, hence $\|\mathcal{D}_H f\|_2 \leq \|f\|_2 + \|\mathcal{M}_H f\|_2 \leq 2\|f\|_2$. Since $\mathcal{D}_H = I - \mathcal{M}_H$ and $\|\mathcal{M}_H\|_{2 \rightarrow 2} \leq 1$, the sharper bound $\|\mathcal{D}_H f\|_2 \leq \|f\|_2$ follows by viewing $\mathcal{M}_H$ as convolution by a unit-mass kernel (Fourier multiplier bounded by 1). For the derivative, note $(\mathcal{M}_H f)' = \mathcal{M}_H(f')$ since convolution commutes with ∂_t , so the same bound applies to f' . $\square$

4 Deterministic low-frequency annihilation from DC-kill

Let $\Gamma(t_0) = \{\gamma : |\gamma - t_0| \leq H\}$ be a local packet of ordinates, and let coefficients b_{γ} satisfy the DC-kill condition

$$\sum_{\gamma \in \Gamma(t_0)} b_{\gamma} = 0.$$

Define the shifted exponential sum

$$S_{t_0}(\omega) := \sum_{\gamma \in \Gamma(t_0)} b_{\gamma} e^{-i\omega(\gamma - t_0)}, \quad F_T(t_0) := \int_{|\omega| \leq 1/T} |S_{t_0}(\omega)|^2 d\omega.$$

Lemma 2 (Taylor annihilation under DC-kill). *For every $T \geq 1$,*

$$F_T(t_0) \leq \frac{2}{3} T^{-3} \left(\sum_{\gamma \in \Gamma(t_0)} |b_{\gamma}| |\gamma - t_0| \right)^2.$$

In particular, if $|\gamma - t_0| \leq H$ for all $\gamma \in \Gamma(t_0)$, then

$$F_T(t_0) \leq \frac{2}{3} H^2 \|b\|_{\ell^1(\Gamma(t_0))}^2 T^{-3}.$$

Proof. Using $\sum b_\gamma = 0$,

$$S_{t_0}(\omega) = \sum_{\gamma} b_{\gamma} (e^{-i\omega(\gamma-t_0)} - 1).$$

Since $|e^{-ix} - 1| \leq |x|$, we get

$$|S_{t_0}(\omega)| \leq |\omega| \sum_{\gamma} |b_{\gamma}| |\gamma - t_0|.$$

Square and integrate: $\int_{-1/T}^{1/T} \omega^2 d\omega = \frac{2}{3} T^{-3}$. □

5 The analytic bottleneck: an explicit safe-line transfer identity

The reduction of RH hinges on controlling the low-frequency energy of the *actual* filtered boundary channel $g_H(t)$. The deterministic Lemma 2 shows that once a DC-kill (dipole) condition is enforced and the packet is localized, low-frequency energy decays polynomially in T . The remaining problem is to connect g_H to a safe-line object for which unconditional mean-square bounds are known.

Theorem 1 (Safe-line transfer identity (missing analytic bridge)). *Fix $\varepsilon, \delta > 0$ and let $g_H(t)$ be the dipole-filtered, mirror-odd, smoothed boundary derivative defined above at $\sigma = \varepsilon$. There exist:*

- *an explicit bounded linear operator $\mathcal{T}_{\varepsilon, \delta}$ (given by a Poisson/Cauchy-type kernel and the functional equation),*
- *a “safe-line” input $f(t) := \zeta'/\zeta(\frac{1}{2} + \varepsilon + it)$ (or a closely related xi-derivative),*
- *and an explicit remainder $r(t)$ (Gamma/background terms),*

such that for all large t (in an averaged sense on $[T, 2T]$),

$$g_H(t) = \mathcal{T}_{\varepsilon, \delta}(f)(t) + r(t),$$

with operator bound

$$\|\mathcal{T}_{\varepsilon, \delta} h\|_{L^2([T, 2T])} \leq C_{\varepsilon, \delta} \|h\|_{L^2([T, 2T])},$$

and remainder bound

$$\|r\|_{L^2([T, 2T])}^2 \ll_{\varepsilon, \delta, w, W} T(\log T)^C.$$

6 Bridge via safe-line mean square (contradiction template)

Assuming Theorem 1, the desired L^2 Bridge bound follows from classical mean-square bounds on the safe line.

Theorem 2 (Bridge from safe-line mean square). *Assume Theorem 1. Suppose there exists a classical bound (unconditional) of the form*

$$\int_T^{2T} \left| \frac{\zeta'}{\zeta} \left(\frac{1}{2} + \varepsilon + it \right) \right|^2 dt \ll_{\varepsilon} T(\log T)^{C_{\varepsilon}}.$$

Then the dipole-filtered boundary energy satisfies

$$\int_T^{2T} |g_H(t)|^2 dt \ll_{\varepsilon, \delta, w, W} T(\log T)^C$$

for some $C > 0$. In particular, failure of the Bridge L^2 energy control would contradict the safe-line mean-square theorem.

Proof. By Theorem 1 and the triangle inequality,

$$\|g_H\|_{L^2([T, 2T])} \leq \|\mathcal{T}_{\varepsilon, \delta} f\|_2 + \|r\|_2 \leq C_{\varepsilon, \delta} \|f\|_2 + \|r\|_2,$$

and the stated safe-line mean square plus the remainder bound yields the claim. $\square$

7 Numerical certification in the ordinates-only model

We certify a low-frequency depletion phenomenon in a reduced model depending only on ordinates.

Data and window

We use a tabulation of high ordinates γ of nontrivial zeros of $\xi(s)$ stored as a single-column file (e.g. Odlyzko-style). Fix a center height t_0 and restrict to a local window

$$\Gamma(t_0) = \{\gamma : |\gamma - t_0| \leq H\}, \quad N = |\Gamma(t_0)|.$$

Smooth dipole (weighted DC-kill)

Let $w_H(u) = H^{-1}w(u/H)$ (Gaussian in experiments). Define the weighted mean

$$\bar{\gamma}_w := \frac{\sum_{\gamma \in \Gamma(t_0)} w_H(\gamma - t_0) \gamma}{\sum_{\gamma \in \Gamma(t_0)} w_H(\gamma - t_0)},$$

and coefficients

$$b_\gamma := w_H(\gamma - t_0) (\gamma - \bar{\gamma}_w).$$

Then $\sum_{\gamma \in \Gamma(t_0)} b_\gamma = 0$ up to floating-point error.

Low-frequency energy

For each $T \geq 1$ compute

$$F_T(t_0) := \int_{|\omega| \leq 1/T} \left| \sum_{\gamma \in \Gamma(t_0)} b_\gamma e^{-i\omega(\gamma - t_0)} \right|^2 d\omega$$

via a uniform ω -grid and trapezoidal quadrature (shifted by t_0 to remove spurious scale constants).

Observed slopes

We estimate $\hat{\beta}$ by a log-log fit:

$$\log F_T \approx -\hat{\beta} \log T + \text{const.}$$

Representative run at height $t_0 \approx 2.6765 \times 10^{11}$ with $N = 10^4$ zeros gives

$$\hat{\beta} \approx 1.7858, \quad \hat{\beta} - 1 \approx 0.7858,$$

and on the upper half of the T -range,

$$\hat{\beta} \approx 2.9729, \quad \hat{\beta} - 1 \approx 1.9729.$$

The DC-kill residual $\sum b_\gamma$ is observed at machine precision ($\sim 10^{-10}$ or smaller).

Interpretation

This numerically certifies a strong low-frequency depletion in the dipole-filtered local packet of ordinates. It does not, by itself, prove Theorem 1; rather, it provides evidence that the designed odd/dipole filtering produces a genuine spectral gap with substantial safety margin at large heights.